

OSCE-III

Respiratory

Asthma

Asthma predictive index

- major
 - Parental asthma
 - Atopic dermatitis/
eczema
 - Inhalational sensitization
- Minor
 - Allergic rhinitis
 - Wheeze apart from cold
 - >4% peripheral blood
eosinophilia
 - Food allergies

- DX

Table 138-6 -- LUNG FUNCTION ABNORMALITIES IN ASTHMA

Spirometry (in clinic):

Airflow limitation:

Low FEV₁ (relative to percentage of predicted norms)

FEV₁/FVC ratio <0.80

Bronchodilator response (to inhaled β-agonist):

Improvement in FEV₁ ≥12% and
≥200 mL*

Exercise challenge:

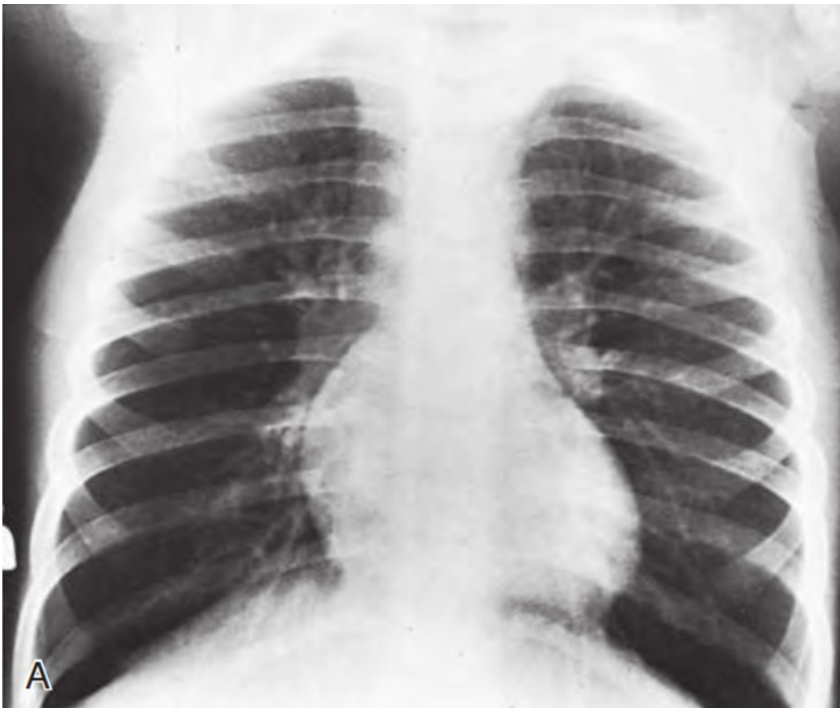
Worsening in FEV₁ ≥15%*

Daily peak flow or FEV₁ monitoring: day to day and/or AM-to-PM variation
≥20%*

Cont'd

- **CXR**

- Hyperinflation
- Atelectasis
- Pneumo mediastinum
- Pneumo thorax



Cont'd

- DDX

Age	Common	Uncommon	Rare
Less than 6 months	Bronchiolitis Gastro-esophageal reflux	Aspiration pneumonia Bronchopulmonary dysplasia Congestive heart failure Cystic fibrosis	Asthma Foreign body aspiration
6 months - 2 years	Bronchiolitis Foreign body aspiration	Aspiration pneumonia Asthma Bronchopulmonary dysplasia Cystic fibrosis Gastro-esophageal reflux	Congestive heart failure
2 - 5 years	Asthma Foreign body aspiration	Cystic fibrosis Gastro-esophageal reflux Viral pneumonia	Aspiration pneumonia Bronchiolitis Congestive heart failure Gastro-esophageal reflux

Severity ass't at emergency

Table 144-4 Formal Evaluation of Asthma Exacerbation Severity in the Urgent or Emergency Care Setting*

	MILD	MODERATE	SEVERE	SUBSET: RESPIRATORY ARREST IMMINENT
SYMPTOMS				
Breathlessness	While walking	While at rest (infant—softer, shorter cry, difficulty feeding)	While at rest (infant—stops feeding)	
Talks in Alertness	Can lie down Sentences May be agitated	Prefers sitting Phrases Usually agitated	Sits upright Words Usually agitated	Drowsy or confused
SIGNS				
Respiratory rate [†]	Increased	Increased	Often >30 breaths/min	Paradoxical thoracoabdominal movement
Use of accessory muscles; suprasternal retractions	Usually not	Commonly	Usually	Absence of wheeze
Wheeze	Moderate; often only end-expiratory	Loud; throughout exhalation	Usually loud; throughout inhalation and exhalation	Bradycardia
Pulse rate (beats/min) [‡]	<100	100-120	>120	Absence suggests respiratory muscle fatigue
Pulsus paradoxus	Absent <10 mm Hg	May be present 10-25 mm Hg	Often present >25 mm Hg (adult) 20-40 mm Hg (child)	

Severity ass't for pt not on long term medication

Table 144-8 Assessing Asthma Severity and Initiating Treatment for Patients Who Are Not Currently Taking Long-Term Control Medications*

	CLASSIFICATION OF ASTHMA SEVERITY			
	Intermittent	PERSISTENT		
		Mild	Moderate	Severe
COMPONENTS OF SEVERITY				
Impairment				
Daytime symptoms	≤2 days/wk	>2 days/wk but not daily	Daily	Throughout the day
Nighttime awakenings:				
Age 0-4 yr	0	1-2x/mo	3-4x/mo	>1x/wk
Age ≥5 yr	≤2x/mo	3-4x/mo	>1x/wk but not nightly	Often 7x/wk
Short-acting β ₂ -agonist use for symptoms (not for prevention of exercise-induced bronchospasm)	≤2 days/wk	>2 days/wk but not daily, and not more than 1x on any day	Daily	Several times per day
Interference with normal activity	None	Minor limitation	Some limitation	Extreme limitation
Lung function:				
FEV ₁ % predicted, age ≥5 yr	Normal FEV ₁ between exacerbations >80% predicted	≥80% predicted	60-80% predicted	<60% predicted
FEV ₁ :FVC ratio [†] :				
Age 5-11 yr	>85%	>80%	75-80%	<75%
Age ≥12 yr	Normal	Normal	Reduced 5%	Reduced >5%
Risk				
Exacerbations requiring systemic corticosteroids:				
Age 0-4 yr	0-1/yr (see notes)	≥2 exacerbations in 6 mo requiring systemic corticosteroids or ≥4 wheezing episodes/yr lasting >1 day and risk factors for persistent asthma		
Age ≥ 5 yr	0-1/yr (see notes)	≥2/yr (see notes)	≥2/yr (see notes)	≥2/yr (see notes)
Consider severity and interval since last exacerbation.				
Frequency and severity may fluctuate over time for patients in any severity category.				
Relative annual risk of exacerbations may be related to FEV ₁ .				

Pulmonary index score (PIS)

- score of 7 to 11: moderate severity
- ≥ 12 indicates a severe attack

Score	Respiratory rate*	Wheezing•	Inspiratory/ expiratory ratio	Accessory muscle use	Oxygen saturation
0	≤ 30	None	2:1	None	99-100
1	31-45	End expiration	1:1	+	96-98
2	46-60	Entire expiration	1:2	++	93-95
3	>60	Inspiration and expiration	1:3	+++	<93

Con'd

- **Danger Signs**
 - Silent chest on auscultation
 - Cyanosis
 - Bradicardia,
 - Profound diaphoresis
 - Paradoxical breathing
 - Drowsiness or confusion
 - Agitation
 - Exhaustion

Asthma- MX

- **Quick-Reliever Medications**

- Short-Acting Inhaled β -Agonists
- Anticholinergic Agents
- Systemic Corticosteroids

- **Long-Term Controller Medications**

- Inhaled Corticosteroids
- Long-Acting Inhaled β -Agonists
- Leukotriene-Modifying Agents
- Nonsteroidal Antiinflammatory Agents
- Theophylline
- Anti-Immunoglobulin E (Omalizumab)

Cont'd

RECOMMENDED STEP FOR INITIATING THERAPY				
	Intermittent	Mild	Moderate	Severe
All ages	Step 1	Step 2	Step 3	Step 3
Age 0-4 yr			Step 3, medium-dose ICS option	Step 3, medium-dose ICS option
Age 5-11 yr				or
				Step 4
Age ≥12 yr			Consider a short course of systemic corticosteroids	Consider a short course of systemic corticosteroids
In 2-6 wk, evaluate level of asthma control that is achieved and adjust therapy accordingly. If no clear benefit is observed within 4-6 wk, consider adjusting therapy or alternative diagnoses.				

Table 144-12 Stepwise Approach for Managing Asthma in Children*

AGE	THERAPY†	INTERMITTENT ASTHMA	PERSISTENT ASTHMA: DAILY MEDICATION				
			STEP DOWN if possible (and asthma is well controlled at least 3 months)		ASSESS CONTROL	STEP UP if needed (first check inhaler technique, adherence, environmental control, and comorbid condition)	
		Step 1	Step 2	Step 3	Step 4	Step 5	Step 6
0-4 yr	Preferred	SABA prn	Low-dose ICS	Medium-dose ICS	Medium-dose ICS + either LABA or LTRA	High-dose ICS + either LABA or LTRA	High-dose ICS + either LABA or LTRA and Oral corticosteroid
	Alternative		Cromolyn or montelukast				
5-11 yr	Preferred	SABA prn	Low-dose ICS	Either low-dose ICS ± LABA, LTRA, or theophylline or Medium-dose ICS	Medium-dose ICS + LABA	High-dose ICS + LABA	High-dose ICS + LABA and Oral corticosteroid
	Alternative		Cromolyn, LTRA, nedocromil, or theophylline		Medium-dose ICS + either LTRA or Theophylline	High-dose ICS + either LTRA or Theophylline	High-dose ICS + either LTRA or Theophylline and Oral corticosteroid
≥12 yr	Preferred	SABA prn	Low-dose ICS	Low-dose ICS + LABA or Medium-dose ICS	Medium-dose ICS + LABA	High-dose ICS + LABA and Consider omalizumab for patients with allergies	High-dose ICS + LABA + oral corticosteroid and Consider omalizumab for patients with allergies
	Alternative		Cromolyn, LTRA, nedocromil, or theophylline	Low-dose ICS + LTRA, theophylline, or zileuton	Medium-dose ICS + LTRA, theophylline, or zileuton		

Management of acute exacerbation

1. **ABC of life**
2. **Oxygen:** goal > 95% Spo₂
3. **Give maintenance fluid**
4. **Rapidly acting bronchodilators**
Nebulized or metered dose inhalers
5. **Steroid (oral or iv)**
Oral- prednisolone
IV- hydrocortisone
 - ❖ If mild to moderate: give steroid when there is no improvement with SABA in 1 hr
 - ❖ If moderate to severe: give SABA & steroid immediately

6. Magnesium sulphate

For severe asthma being treated with SABA & steroid

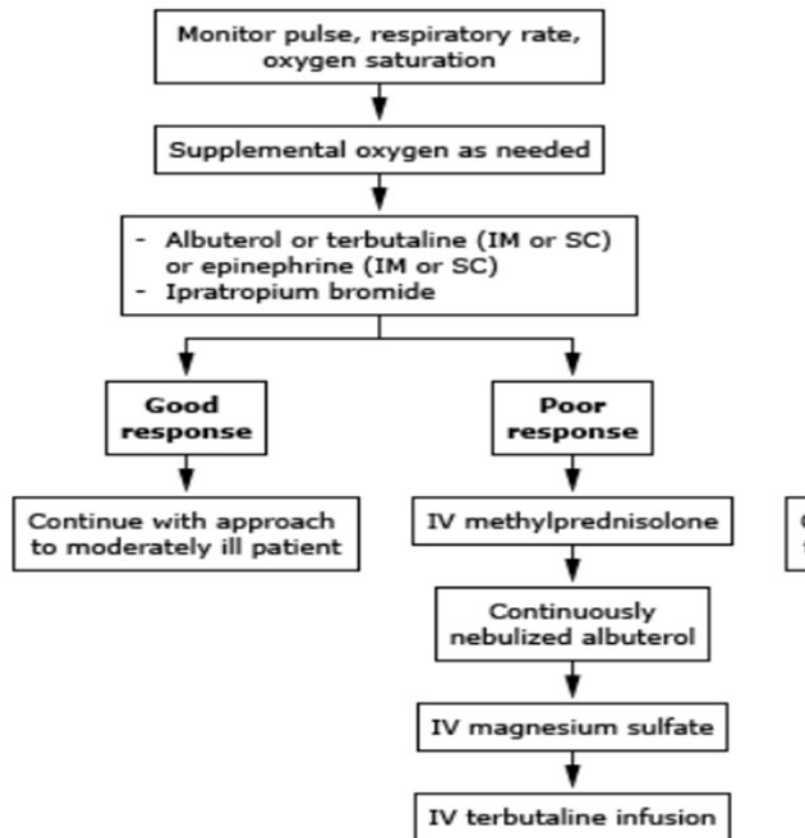
Safer than Aminophylline

7. Aminophylline

Reserved for those who don't respond to treatment

Admit to ICU if available

Management of severe asthma in children



Cont'd

- Monitor
 - Vital Q3 hr
 - If still deteriorating, get CXR
- Follow up
 - Check for improv't & reclassify
 - Asses medication technique, adherence
 - Achieve env'tal control
- Discharge
 - Sustained improv't in sx
 - PEF > 70% of predicted
 - Spo2 > 92% at room air for 4 hr

Pertussis

Clinical case definition

- cough lasting at least two weeks with at least one of the following symptoms:
 - Paroxysms (i.e. fits) of coughing
 - Inspiratory whooping
 - Post-tussive vomiting without other apparent cause

Criteria for laboratory confirmation

- Isolation of *Bordetella pertussis* or
- Detection of genomic sequences by means of the PCR or
- Positive paired serology

Cont'd

- A child with broken blood vessels in eyes and bruising on the face due to pertussis coughing





c/f

1. Catarrhal stage (1-2 wk)

congestion and rhinorrhea, low-grade fever, sneezing, lacrimation, and conjunctival suffusion.

3. Convalescent stage (≥ 2 wk), the number, severity, and duration of episodes of cough diminish.

2. Paroxysmal stage (2-6 wk) :

- Dry, intermittent, irritative hack and evolves into the inexorable paroxysms that are the hallmark of pertussis.
- **A well-appearing toddler begin a machine-gun burst of uninterrupted cough on a single exhalation, chin and chest held forward, tongue protruding maximally, eyes bulging and watering, face purple, until coughing ceases and a loud whoop follows as inspired air traverses the still partially closed airway.**
- Post-tussive emesis followed by exhaustion
- more than 1 episode/hr at the peak of the paroxysmal stage

Dx

DDx

- Protracted/paroxysmal coughing caused by
 - Mycoplasma,
 - Parainfluenza viruses,
 - Influenza viruses,
 - Enteroviruses,
 - Respiratory syncytial viruses (RSV), or adenoviruses.

Investigation

- **WBC:** 15-100,000 cells/ μ l, with predominant lymphocytosis in catarrhal stage, Normal sized lymphocytes (large size in viral infections)
- **Thrombocytosis** :556-782,000/ μ l
- **CXR** : most of the time normal , but may show perihilar infiltrates with butterfly appearance, atelectasis & consolidation if there is secondary bacterial infection
 - Occasionally :
pneumothorax/mediastinum, SC emphysema
- **Culture , PCR and serology** have limited sensitivity & specificity
- **Echo:** PHtn in severely ill patients

Comp.

Complications

- Pneumonia
 - the commnest
 - common Mo.: s.aureus, strep. pneum,
- Otitis media
- Puim. HTn
- Convulsion
- Malnutrition
- Apnea and bradycardia
- CNS abnormalities
- Bronchiectasis
- Forceful cough
 - i. hernia
 - Petechie
 - Cns hemorrhage

- Admission criteria:
 - Infants < 3mo
 - Infants 3-6mo with severe paroxysms
 - Premature babies
 - Children with underlying cardiac, pulmonary, neurologic or muscular disorders
- Non-life threatening paroxysm
 - <45 seconds
 - Red but not blue
 - Tachycardia ,
 - bradycardia if not <60/min
 - Oxygen desaturation resolving spontaneously
 - Whooping or brisk self rescue at the end of paroxysm
 - Self expectorated mucus plug
 - Post tussive exhaustion but not unresponsiveness

Mx

1. ABC of life
2. Decide on admission
3. Supportive care
 - Airway MX
 - oxygen
 - Avoid procedures that trigger cough (NGT)
 - Antipyretics
 - Encourage BF
 - fluid
4. Antibiotic
 - **Azithromycin** 10 mg/kg safe for 5 days for all age groups
 - **Erythromycin** 40-50mg/kg divided in to 4 doses for 14 days for age group above 1 mo
 - **Clarithromycin** 15 mg /ke/day divided in 2 doses for 7 days for > 1mo of age
 - **Cotrimoxazole** (TMP8mg/kg/day_ SMZ 40mg/kg/day divided in 2 doses for 14 days for those above 2 mo of age

- Discharge

- Infant tolerating cough
- Can feed enough to gain weight
- Caretakers are reliable
- Follow up is assured

- Isolation

- Until 5 days of macrolid therapy

Pneumonia

Age and Bacterial Pathogens

Newborns ;

Group B streptococcus, Escherichia coli, Klebsiella species, Enterobacteriaceae

1- 3 months

Chlamydia trachomatis

Preschool

Streptococcus pneumoniae, Haemophilus influenzae type b, Staphylococcal aureus,

- Less common: group A streptococcus, Moraxella catarrhalis, Pseudomonas aeruginosa

School

- Mycoplasma pneumoniae, Chlamydia pneumoniae

Dx

- DX
 - Mainly clinical
 - CRP/ ESR
 - PICT
 - CXR
 - Pleural fluid analysis
 - Blood culture
 - Sr electrolyte
 - Sputum analysis
- Indication for cxr
 - Recurrent pneumonia
 - Complicated
 - No response to mx
 - Other DX likely
 - HIV +ve, sever disease

- CXR
 - h. pneumonia.. tracheo
bronchial
 - s. pneum.... lobar
consolidation
 - GAS.... interstitial
 - s.aureus ...pneumatocle
 - Atypical also pcp...
interstitial infiltrate

Comp

- Local

- Pleural effusion
- Empyema
- Pneumothorax
- Bronchopulm. Fistula
- Lung abscess

- Systemic

- Meningitis
- Pericarditis
- Endocarditis
- Osteomyelitis
- Septic arthritis
- sepsis

Mx

Admission criteria

- Age <3mo
- Multiple lobe involv't
- Immunocompromised
- Toxic appearance
- Moderate- sever resp. distress
- O2 need
- Complicated pneumonia
- DHN
- Inability to tolerate PO
- NO response to PO med
- Social factor

Cont'd

- Out pt
 - Amox. 40-50 mg/kg/ day
TID for 7-10 days or
 - Cotrimoxazole
48mg/kg/day
 - Antipyretics
 - ❖ Azitromycin 10mg/ kg/
day for school age
- In pt
 - ABC of life
 - Oxygen
 - Antibiotic therapy
 - Fluid
 - Nutritional
 - Antipyretics
 - Anti pain

Cont'd

- Inpt (antibiotic therapy)
 - Crystalline penicilline or
 - Ampicillin + gentamycin in immunocompromised sever malnutriton
- second line
 - Ceftriaxone
- Discharge
 - Resp distress resolve
 - No hypoxemia
 - feeding well
 - Completed parentral AB

Bronchiolitis

- DX
 - Clinical
 - a previously healthy infant presenting with a first time wheeze
 - CBC- usually normal
 - CXR
- Admission criteria
 - Apnea
 - SaO₂ < 90%
 - Inability to feed or drink
 - Convulsion
 - Moderate to severe resp . distress

Cont'd

- Mx
 - Oxygen
 - Fluid
 - Antipyrtics
 - Nutrition
 - ❖ Ab is not needed unless pneumonia
- Compilation
 - DHN* commonest cause of death
 - atelectasis
 - Tension pneumothorax

Foreign body

- Larynx – 3 %
- Trachea/carina – 13 %
- Right lung – 60 %
- Left lung – 23 %
- Bilateral – 2 %

DX

- CXR
 - Localized wheeze: hyper inflation best seen on Exp view
 - Absent air entry: atelectasis, mediastinal shift, compensatory hyperinflation of the other lung; best seen with insp view

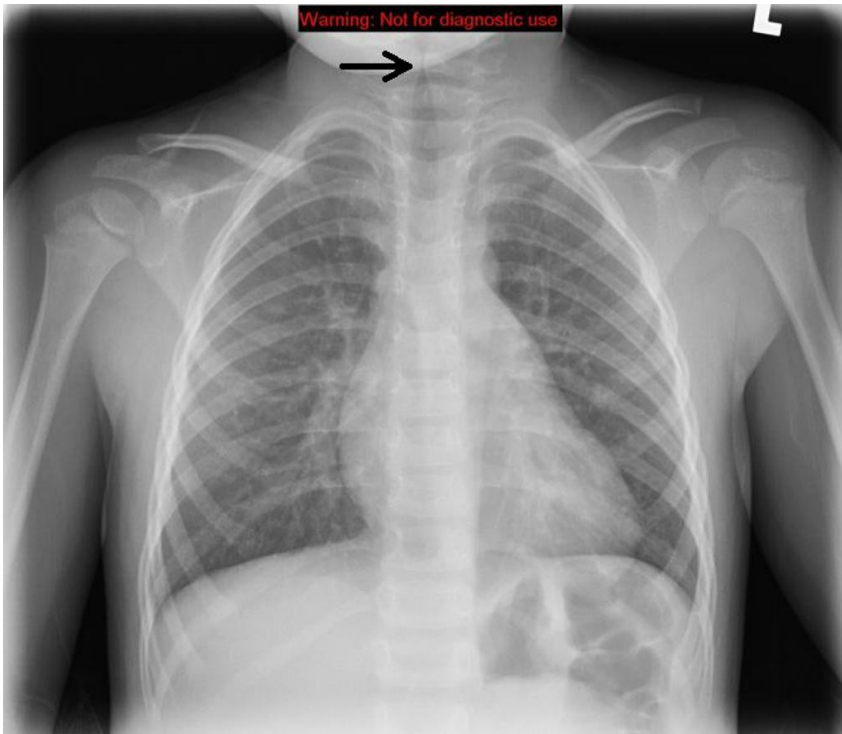
Mx

- <1 year: 5 Back slaps and 5 chest thrusts
- Heimlich Maneuver
- If **rigid bronchoscopy** is unsuccessful, **surgical bronchotomy or segmental resection** may be necessary
- Chronic bronchial obstruction with bronchiectasis and destruction of lung parenchyma may require **segmental or lobar resection**
- **Antibiotics** for secondary infections
- **Steroids** for inflammation
- Treat complications

Croup (laryngotracheobronchitis)

- Cause
 - Rhinovirus
 - RSV
 - Para influenza
- CF
 - Illness often begins with rhinitis
 - Cough - harsh or “barking” and inspiratory stridor
 - Airway obstruction
 - Onset of stridor usually occurs over 6–12 hours

DX



- CXR
 - Steeple sign or inverted pencil sign
- Laryngoscope
 - Erythematous edema with destruction of the epithelium

Level of consciousness

- ☐ Normal, including sleep (0 points)
- ☐ Disoriented (5 points)

Cyanosis

- ☐ None (0 points)
- ☐ With agitation (4 points)
- ☐ At rest (5 points)

Stridor

- ☐ None (0 points)
- ☐ With agitation (1 point)
- ☐ At rest (2 points)

Air entry

- ☐ Normal (0 points)
- ☐ Decreased (1 point)
- ☐ Markedly decreased (2 points)

Retractions

- ☐ None (0 points)
- ☐ Mild (1 point)
- ☐ Moderate (2 points)
- ☐ Severe (3 points)

Westley croup severity score
(appropriate for use in children ≤ 6
years of age)

Score	Severity	Description
0 to 2	Mild	Occasional barking cough, no stridor at rest, mild or no retractions
3 to 7	Moderate	Frequent barking cough, stridor at rest, and mild-to-moderate retractions, but no or little distress or agitation
8 to 11	Severe	Frequent barking cough, stridor at rest, marked retractions, significant distress and agitation
12 to 17	Impending respiratory failure	Depressed level of consciousness, stridor at rest, severe retractions, poor air entry, cyanosis or pallor

mx

- ABC of life
- Supportive care
- Steroid
- Nebulised epinephrine

❖ No need of Ab

Peritonsillar abscess



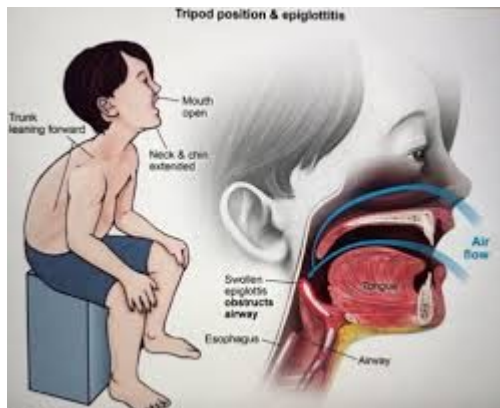
- GAS and mixed anaerobs are the most common pathogens
- Mx
 - Cephalexin+/- metronidazole or
 - Clindamycin
 - Or
 - amoxclavulunate

Epiglottitis

- Cause
 - Hib
 - S. pyogens
 - Pneumococcus
 - S.aureus
- CF
 - High fever, soar throat, dyspnea
 - Swallowing and breathing difficulty
 - Tripoid positioning
 - Drooling of saliva and hyper extended neck
 - Stridor is a late finding



- Tripoid positioning





DX

- CXR
 - Thumb sign on lateral neck xray
- Laryngoscope
 - Large cherry red epiglottitis
- Blood culture

Mx

- ETT or tracheostomy
 - oxygen
- IV Ab
 - Ceftriaxone once daily for 5 days

Stridor DDX

- Cruop
- Epiglottitis
- Laryngeal polyp
- laryngeomalasia
- tracheomalsia

Tuberculosis

6.1 BASIC CONCEPTS OF TB IN CHILDREN

Tuberculosis (TB) is a major public health problem throughout the world. Almost 1.3 million cases and 450,000 deaths occur in children each year. The burden of TB in Ethiopia is one of the highest in the world. Childhood TB reflects recent transmission within a community. Cases of TB in children usually represents between 5-15% of all TB cases.

Peculiarities of childhood TB:

- The commonest age of presentation of childhood TB disease is between 1 and 4 years.
- Young age is a risk factor for infection, for progression from infection to disease, and for spread of disease to other parts of the body, i.e. dissemination.
- Risk of progression to disease is increased when primary infection occurs in the very young (0–4 years) – and in immuno-compromised children.
- Children under one year of age are more liable to develop Miliary and TB meningitis
- Most children with TB are not infectious to others.
- The commonest type of TB in children is smear-negative PTB.
- In adolescents, PTB is generally like adult PTB

Children at greater risk of developing TB include:

- Children who are in close contact with a newly diagnosed smear-positive TB case
- Children less than 5 years of age
- HIV-infected children
- Severely malnourished children

Age influence on the Disease progression: Young children carry higher risk of disease progression to Active TB disease right after recent exposure to infectious cases. The risk peaks among under-five children as the immune system that prevent disease progression

6.1.2 Clinical manifestations of TB in children

Clinical Manifestations: In children TB disease presents in various clinical forms: Primary or post-primary pulmonary tuberculosis; Acute disseminated tuberculosis (meningitis & Miliary tuberculosis); and Extra pulmonary tuberculosis.

Signs and symptoms of childhood TB:

The clinical features of childhood TB are constitutional symptoms and local signs and symptoms which depend on the part of the body affected.

Pulmonary Tuberculosis: The most common clinical presentation is persistent respiratory symptoms and poor weight gain. A child may have non-productive cough and /or mild wheezes.

In infants or HIV-infected, pulmonary TB can also present as acute pneumonia.

More than 50% of infants and children with radiographically moderate to severe pulmonary tuberculosis have no physical findings and are discovered only by contact tracing.

Non-productive cough and mild dyspnea are the most common symptoms.

Systemic complaints such as fever, night sweats, anorexia, and decreased activity occur less often.

Some infants and young children with bronchial obstruction have localized wheezing or decreased breath sounds that may be accompanied by tachypnea or, rarely, respiratory distress.

Extra-pulmonary tuberculosis in children: Extra-pulmonary TB (EPTB) can occur at any age. Young children are particularly susceptible. Children of less than 2 years of age are at risk of disseminated disease causing Miliary TB or TB meningitis. If a patient has extra-pulmonary TB, look for pulmonary TB.

6.2 TB CASE FINDING IN CHILDREN

In addition to the case finding approaches used for adults, integrating TB screening services in to child health services(IMNCI/ICCM and at especial service clinics) are important intensify TB case finding in children:

Systematic TB screening at IMNCI/ICCM clinic: At IMNCI/ICCM clinic, Health workers should consider possibility of TB in a sick child at initial visit and/or during follow up visit:

i) Consider TB in a sick child at first visit:

- Presumptive TB should be considered and evaluation for TB should be initiated, if the child has either one or more of clinical manifestations consistent with TB in children.
- Besides, for under-five child with contact history with infectious TB patients, appropriate evaluation for TB should be made and if Active TB is excluded, treatment for latent TB should be administered.

ii) Consider TB in a child during follow up visit:

If a sick child who initially presented with cough is not responding to either to first line antibiotic treatment for pneumonia or to one-month of standard nutritional therapy for malnutrition, or has recurring pneumonia and has confirmed HIV infection.

At the community level, health extension workers additionally should identify and refer a sick or TB exposed child during regular household visit and upon identification of an index TB case for appropriate evaluation.

6.3 DIAGNOSIS OF TB IN CHILDREN

Health care workers should try to diagnose TB in children at outpatient settings based on careful clinical assessment. Contact history is a very important part of assessment for children suspected of having TB. As young children rarely produce sputum, bacteriologic confirmation TB in children relies on clinical evidences. However, the health care worker should attempt to do gastric aspiration to get adequate material for smear examination. For older children capable of expectorating, sputum sample is to be collected as for adults.

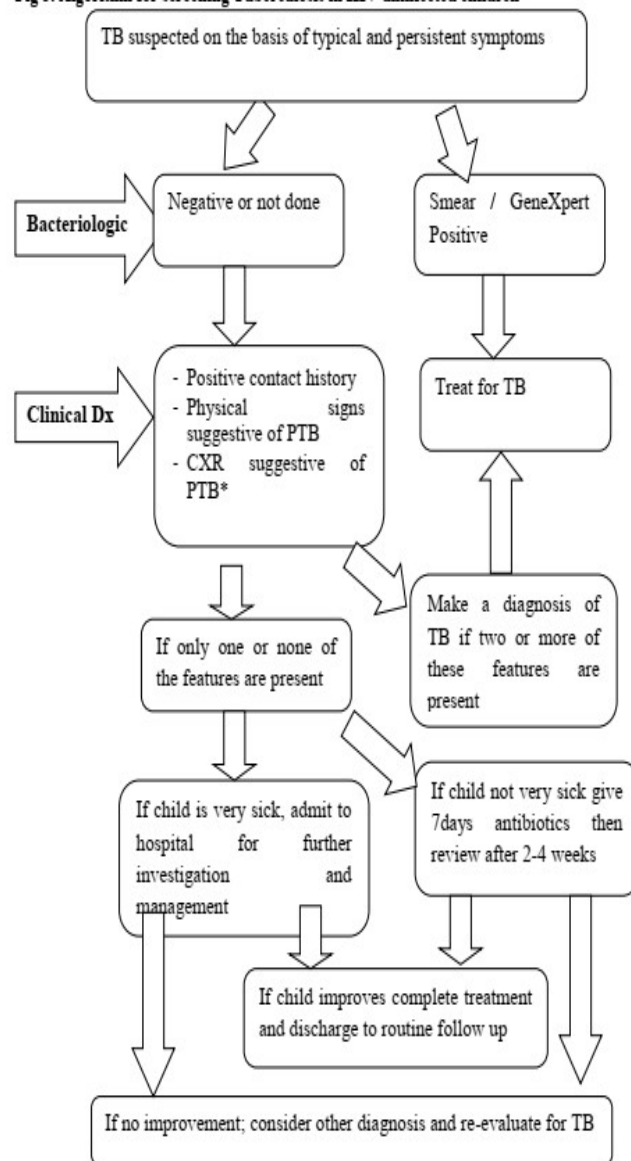
The diagnosis of TB can be made with confidence in the majority of children

using careful clinical assessment.

There are two key factors in suspecting tuberculosis in children:

- (1) Identification of an infectious adult close to the child, especially in the family, and;
- (2) Loss of weight or failure to thrive.

Fig 5. Algorithm for screening Tuberculosis in HIV uninfected children



Recommended TB regimen for children diagnosed with TB

TB Patient type	Standard Regimen		TB types receiving the regimen
	Intensive phase	Continuation phase	
Drug susceptible TB case	2(RHZE)	4(RH)	• All pulmonary TB cases • Most Extrapulmonary TB cases
	2(RHZE)	10 (RH)	• Extrapulmonary TB patients involving: ▪ CNS (meningitis, tuberculoma) ▪ vertebra and Osteoarticular space
Drug Resistant TB(at least to rifampicin)	Second line anti-TB drugs		• Confirmed or clinically diagnosed DR-TB cases

* previously treated TB patients are treated with the same regimen as for New TB patients for six months after ruling out the presence of resistance at least for Rifampicin with rapid DST.

Additional notes of treatment of TB in children:

1. Administer Anti-TB drugs daily.
2. As children approach a body of 25 kg, Adult dosage can be used
3. Children receiving treatment must be weighed at least every month and adjust the dose
4. Treatment doses should be adjusted as soon as a child changes weight bands
5. Check tablet strengths regularly to avoid toxicity
6. Pyridoxine is recommended for children with severe malnutrition, or taking ART
7. Ethambutol is safe in children at a dose of 20 mg/kg (range 15– 25 mg/kg) daily.
8. Adverse events are less common in children than in adults.

Corticosteroids: Corticosteroids may be used for the management of some complicated forms of TB, e.g. TB meningitis, airway obstruction by TB lymph glands, and pericardial TB (for dosing see treatment of severe forms of TB in adults).

4.2 TB CASE DEFINITIONS

The programmatic definitions Tuberculosis are based on the level of certainty of the diagnosis and on whether or not laboratory confirmation is available.

Presumptive Tuberculosis: Refers to a patient who presents with symptoms or signs suggestive of TB, in particular cough of two weeks or more duration.

Bacteriologically confirmed TB case: Refers to a patient from who has at least one positive result either by smear microscopy, culture or Xpert MTB/RIF assay.

Clinically diagnosed TB case: Refers to a patient who does not fulfill the criteria for bacteriological confirmation but has been diagnosed with active TB by a clinician and decided to treat with a full course of TB treatment.

4.3.3 Classifications based on drug resistance

Due to the increasing burden of drug resistant TB, susceptibility to the core Anti-TB drugs in particular to Rifampicin is the current main criteria to design the likely effective TB treatment regimen.

Note that Selection of TB treatment regimen does not consider information from bacteriological confirmation, site of disease and HIV status of the patient.

- a) **Mono-drug resistance:** resistance to one first-line anti-TB drug only.
- b) **Poly-drug resistance:** resistance to more than one first-line anti-TB drug (other than both Isoniazid and Rifampicin).
- c) **Multidrug resistance (MDR-TB):** resistance to at least both Isoniazid and Rifampicin.
- d) **Extensive drug resistance (XDR-TB):** resistance to any fluoroquinolone and to at least one of three second-line injectable drugs (Capreomycin, Kanamycin and Amikacin), in addition to multidrug resistance (- Isoniazid and Rifampicin).
- e) **Rifampicin resistance (RR-TB):** resistance to Rifampicin detected using phenotypic or genotypic methods, with or without resistance to other anti-TB drugs. It includes any resistance to Rifampicin, whether mono-drug resistance, multi-drug resistance, poly-drug resistance or extensive drug resistance.

5.5 TREATMENT OF SERIOUS FORMS OF EXTRAPULMONARY TB

The basic principles that underlay the treatment of pulmonary TB also apply to treatment of TB in Extrapulmonary organs. However, patients with serious form of ETB (involving

pericardium or meninges) should be managed at hospital level with the help of specialist as they are usually associated with fatal outcome.

Recommended additional actions for patients with Extrapulmonary TB include:

- ☞ Extend the continuation phase of treatment to 10 months for TB involving Vertebra, Osteoarticular space, or CNS (:meningitis or tuberculoma): 2(RHZE)/10RH
- ☞ Administer adjuvant treatment with corticosteroids for severe form of TB including:
 - CNS TB
 - Pericardial TB
 - Adrenal TB
 - Miliary TB with respiratory distress especially in children
 - Vertebral TB with signs of weakness in the extremities.

Administration of steroid as adjuvant treatment:

- Administer prednisolone 1mg/kg (Max 60mg/day) for total period of six to eight weeks
- Start with 60 mg of prednisone for 4 weeks, and taper progressively over the next weeks.

Or,

- Administer dexamethasone for a total of 6-8 weeks.
- Start with initial dose of 8 mg per day for children < 25 kg and 12 mg per day for children > 25 kg and adults for 3 weeks and then taper the dose during the following 3 weeks.

3.2.3 Treatment of latent TB infection

Treatment of latent TB infection benefits those population groups known to have higher risk of progression to Active TB disease if infected.

The recommendation of treatment of latent TB infection in Ethiopia includes:

- ❖ Under-five children who are exposed to TB in the past one year, &
- ❖ People living with HIV regardless of their age

Isoniazid Preventive Therapy (IPT): isoniazid is given to individuals with latent infection with *Mycobacterium tuberculosis* in order to sterilize the infection and hence, prevent progression to active disease. Screening for exclusion of active TB in HIV infected persons is the single most important step that should precede the decision to initiate IPT.

Studies have shown that providing IPT to treat latent TB infection does not increase the risk of developing INH-resistant TB as long as active TB is rule out. Therefore, concerns regarding the development of INH resistance should not be a barrier for IPT provision.

IPT for HIV infected population: should be provided to all HIV-infected individuals who are unlikely to have active TB irrespective of their CD4 count, ART status, pregnancy status or history of treatment for prior episode of TB before three years.

IPT for HIV negative TB exposed under-five children: should be administered for under-five asymptomatic children who are exposed to TB within the past one year.

Dosing of INH for IPT:

The dose of INH is 300mg/day for adults and 10mg/kg for children daily administered for six months. It is also advised to co-administered vitamin B-6 (25mg/day) with Isoniazid to prevent INH-induced peripheral neuropathy.

Table 4: showing weight based dosing of INH preventive therapy for children

Weight Ranges(kg)	Dose Given (mg)	INH 100mg tab
< 5	50	½ tablet
5.1-9.9	100	1 tablet
10-13.9	150	1 ½ tablet (or ½ adult tablet)
14 -19.9	200	2 tablets
20 -24.9	250	2 ½ tablets
>25	300	1 adult tablet

Contraindications to IPT

Individuals with any one or more of the following conditions should *not* receive IPT.

- Symptoms compatible with tuberculosis even if the diagnosis isn't yet confirmed.
- Active hepatitis (chronic or acute)
- Regular and heavy alcohol consumptions
- Prior allergy or intolerance to isoniazid
- Symptoms of peripheral neuropathy

Note that past history of TB and current pregnancy should not be contraindications for administering IPT.

NB: Administer IPT for TB exposed asymptomatic under five children when the index cases has pulmonary TB (both Bacteriologically and clinically diagnosed).

Placement of IPT Clients

IPT should be part of a comprehensive care for HIV positive individuals; therefore, these patients should be initiated and monitored at ART clinic and the information should be documented in the Pre-ART/ART register.

IPT for HIV negative under-five children should be administered and followed up at the TB clinic and information should be recorded on unit-TB register.

Monitoring of Patients on IPT

Patients should be given one-month supply of Isoniazid for six months. At each follow-up visit, the health care worker should:

- Educate patient about, adherence side effects, and importance of coming to health facility if develops symptoms suggestive of TB.
- Evaluate and counsel patients on importance of adherence to treatment
- Evaluate for drug toxicity including signs/symptoms of hepatitis, peripheral neuropathy, and rash).
- Evaluate for signs and symptoms of active tuberculosis or other OIs.
- Stop IPT, if active TB is diagnosed and start full course of anti-TB treatment.

Adherence monitoring on IPT: patients should be supported at home level either by HEWs or family supporter to ensure daily administration Isoniazid. Monthly scheduled follow up needs to be integrated with other treatment services for which the client is appointed for.

Treatment interruption management: IPT is said to be completed if a patient completed the full course of therapy within nine months period (i.e. the six months doses should be finished in nine months' time).

If the client discontinues treatment for a period of less than three months:

- Resume the same course by adding for the missed doses at the end

If the client discontinues treatment for a period of more than three months:

- Re-initiate new course of IPT for six months

Hence, if a patient has interrupted isoniazid preventive therapy without the medical personnel advice, the client should be traced by HEW or through the index person, and treatment must be resumed after identifying and addressing the adherence barriers.

National policy on use of IPT for HIV infected patients:

- *IPT should be administered at enrolment to HIV care after ruling out active TB*
- *IPT is to be administered once and should not be repeated unless there is strong indication on its benefits which is to be decided by senior physician*
- *IPT should be administered only for six months*
- *IPT should not be administered right after completing full course of TB treatment*
- *IPT can be administered for patients who had history of TB treatment before three years*

Differential diagnosis of chronic cough

- TB
- Hyperactive airway disease
- Lung abscess
- CHF
- Pertusis
- Chronic bronchitis..rare
- Contact hx, around 95% will develop disease within 2 years of contact
- Gastric aspirate
 - NPO FOR 4-6 HRS
 - 30ML NS via NG tube
 - Aspirate after 10-15 minutes

- Primary TB is common in pediatric age group
- Know when to give steroid
- IPT
- Scrofula or TB lymphadenitis is the most common form of extrapulmonary TB

TB meningitis

1st stage

- Non specific symptoms
 - Fever, headache, irritability, drowsiness, malaise
- Focal neurological signs absent
- Stagnation or loss of developmental milestone

2nd stage

- More abrupt
- Most common features
- Lethargy, nuchal rigidity, seizures, positive PE, vomiting, hypertonia, CN palsy, Focal neurologic signs
- Hydrocephalus
- Raised ICP
- Vasculitis

- 3rd stage
- Coma
- Hemi or para plegia
- Hypertension
- Decerebrate posturing
- Deterioration of vital signs
- Death

- Prognosis
- 1st excellent
- 3rd Permanent disability

Diagnosis of TB(pul+others)

- Key-high index of suspicion
- AFB microscopy

Ziehl-Neelsen's stain



2.2.1 Number of sputum specimens and collection schedule for AFB microscopy and Xpert MTB/RIF assay

Revised National recommendations on the use of smear microscopy for the diagnosis of Tuberculosis:

- ❖ *The recommended number of sputum specimen required for evaluation using microscopy is changed from three samples (spot-morning-spot) to two samples (spot-spot).*
- ❖ *One sputum smear positive result confirms the diagnosis of bacteriologically confirmed Tuberculosis.*
- ❖ *Considering the increased sensitivity of Xpert MTB/RIF Assay over Sputum smear microscopy to diagnose TB; Xpert MTB/RIF Assay is preferred initial test for patients who are children and/or people living with HIV (PLHIV).*
- ❖ *For patients whose sputum smears microscopy result is twice negative and in whom clinical suspicion for TB remains high after administration of Broad spectrum Antibiotic, Xpert MTB/RIF Assay is advised as follow on test.*

Collect two "on-the-spot" sputum samples for smear microscopy:

Instruct to produce the sample on same day, give one labelled container to produce the first sample immediately, and give the second container when the patient brings the first sample and instruct to produce the second sample after 30 minutes to 1 hour.

Collect one "on-the-spot" sputum samples for Xpert MTB/RIF Assay:

Instruct the patient on the techniques of quality sample production, and Give one labelled falcon tube to Collect the sample.

Why the change from three to two consecutive sputum specimens (spot-spot) schedule?

Answer: Based on evidences, good quality microscopy of two consecutive sputum specimens (spot-spot) identifies 95–98% of smear-positive TB patients in quality Assured laboratories.

Advantages:

- Greatly reduces the workload of laboratories
- Has the potential for offering same-day diagnosis
- Is better for patients because it reduces the number of visits while largely maintaining sensitivity.

Disadvantages:

- Very minor loss in the number of cases detected.

Cult

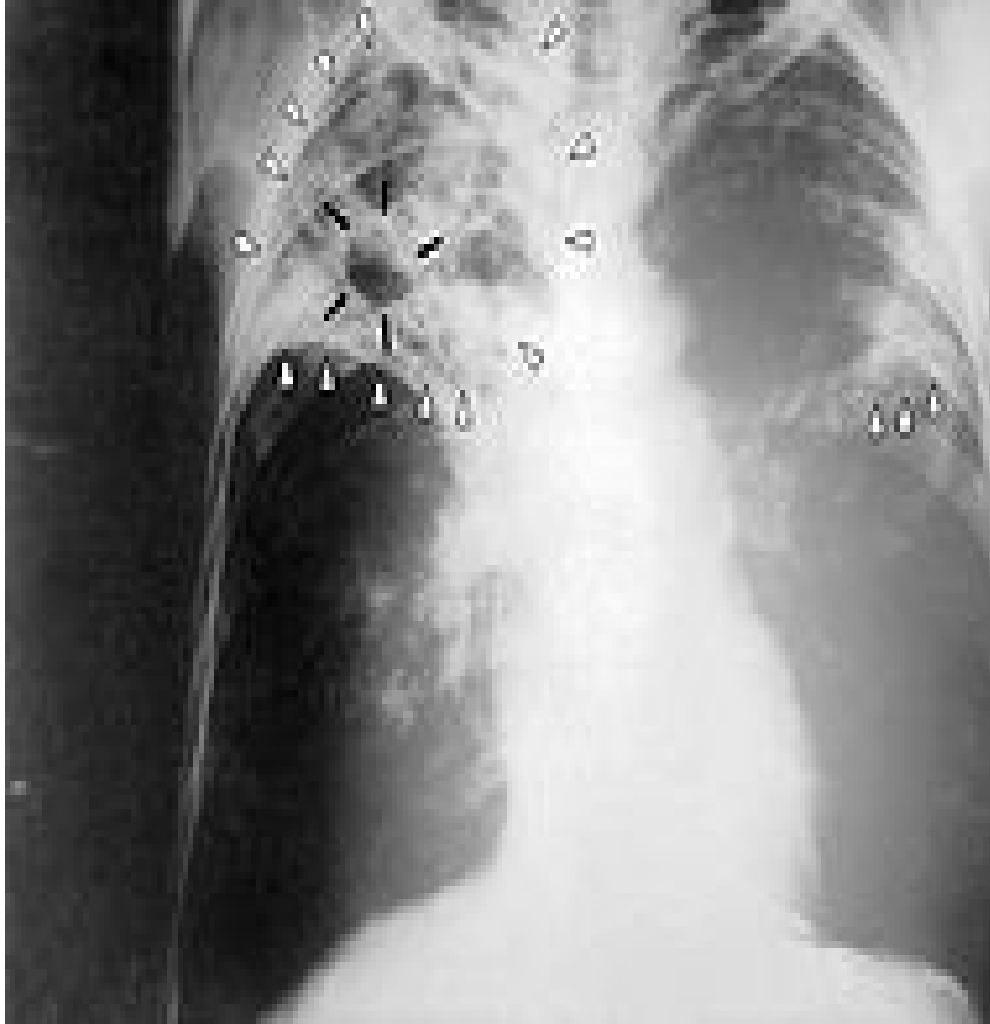
- Culture-sputum/
other specimen
 - **Lowenstein-Jensen
or Middle
brook media**(takes 4-
8 wk for growth)-
popular
 - Liquid media-2-3 wk
for growth-by PCR or
paper
chromatography
- *Solid media/ Liquid
media*



Diagnosis of TB(pul+others)

- Neuclic acid amplification
 - Within hours-lower sensitivity/high cost
 - To confirm-smear +ve or for smear _ve or EPTB
 - LPA
 - Gene Xpert
- Drug susceptibility test
 - Relapse,failure cases
 - Media (solid-slow,fluid-fast)
 - PCR-mutation-markers
- Radiography
 - “Classic”-upper lobe infiltrate/cavity,post primary,lower lobe/hilar Lap-primary/atypical

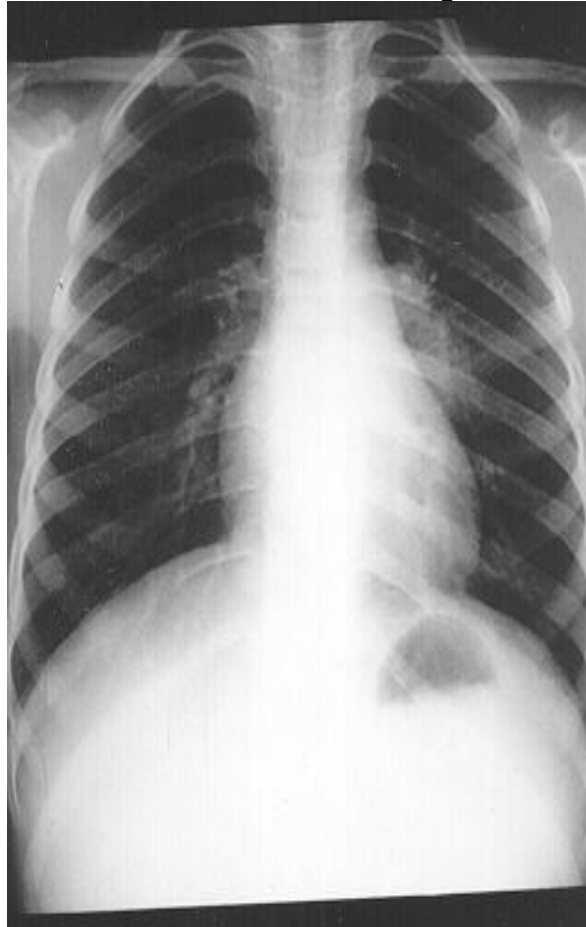
Fibrotic- cavity



Primary tuberculosis: A 31-year-old woman from Taiwan with mild cough and fever. There is an infiltrate in the right lower lobe. PPD was positive. Primary TB must be considered in the differential of this radiograph. (*Courtesy of Wallace Miller, Jr.,*



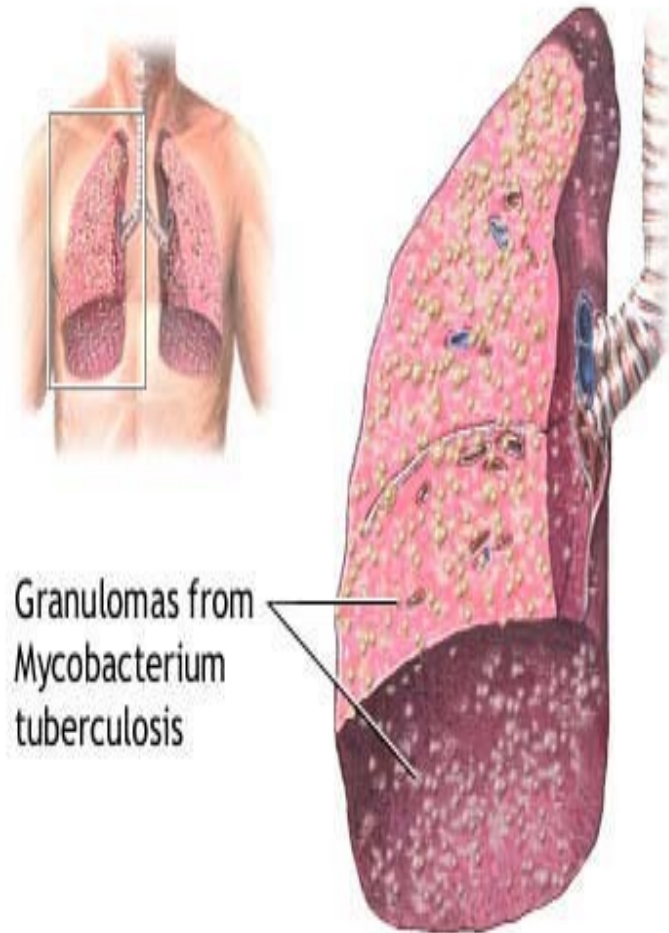
Primary tb



Reactivation tuberculosis: A 31-year-old man with persistent right chest pain. A. View of the lung apices demonstrates an indistinct pulmonary opacity in the left apex. Sputum cultures were positive for *M. tuberculosis*



Milliary tb



Diagnosis of TB(pul+others

- PPD skin test
 - For screening test for M.tbc infection
 - Limited value for diagnosis b/c low sensitivity/specificity
- Additional diagnostic procedures
 - Smear/ culture on
 - Sputum induction-dry cough
 - Bronchoscopic BAL/transbronchial biopsy
 - Gastric lavage-children(early morning)
- EPTB
 - CSF analysis, culture
 - Body fluids: analysis/smear/culture/ADA
 - Tissues: Pleural tissue/LN
 - Bone marrow-miliary
 - Liver biopsy/culture-high in HIV
 - Urine-AFB on culture _ve pyuria ,IVP
 - U/S,C/T/MRI

Principles of management

1. Chemotherapy

Act on different
population of the bacteria

2. Response

Side effect

Appetite

Weight gain

3. Rehabilitation

Malnutrition

4. Screening

Children < 5

Immunosuppression

> 5 years symptomatic

Do investigation...if
positive treat for TB if not
INH prophylaxis for 6
months

TREATMENT OF TB

- Chemotherapy
 - 1.cure of the patients
 - 2.prevention of death from active TB and late effect
 - 3.avoid relapse or recurrence
 - 4.prevention of spread of drug resistance organism
 - 5.protection of the community
- First line essential drugs
 - bactericidal-R,H,Z,S
 - bacteriostatic-E,T
- Second line drugs
- The bacteria population consists of
 - 1.metabolically active,continuously growing bacteria which are in walls of tuberculous cavity
 - 2.intracellular bacilli
 - 3.semiodormant bacteria which undergo spurts of metabolism and
 - 4.dormant bacilli which die off gradually on their own
- Mechanism of action
 - INH-most potent bactericidal
 - Kills 90% bacilli in few days
 - Most active against metabolical active/growing m.o
 - Rifa
 - Good bactericidal(for semiodormant)good sterilizing agent-prevent relapse
 - PZA
 - Kills intracellular mo
 - Sterlizing agent
 - For initial phase-no importance beyond 2 months

Principles of antiTB chemotherapy

- Combination-
 - ↓resistance(mutants will be killed by one of the drugs)
 - Bacteria of different population
- Given for several months(>6 months)
 - To kill slow growing organisms/semi dormant bacilli
 - To prevent relapse
- Resistance low probability-even lower when drugs combined
 - MDR
 - XDR
- Categorization of TB patients by outcome of recent treatment
 - New: never treated with anti TB or <1 month treatment
 - Relapse: previously treated /completed treatment/smear /culture +ve TB
 - Failure: newly diagnosed TB smear +ve at 5th month of treatment
 - Return after default: patient come after discontinuation of drugs for 2 month smear +ve

Otitis media

Otitis media

- 2 main categories:
 - acute otitis media (AOM)
 - otitis media with effusion (OME)
- Peak during the 1st 2 yrs of life.
- Leading reason for physician visits and for use of antibiotics
- Figures importantly in the differential diagnosis of fever
- Accurate diagnosis of AOM in infants and young children may be difficult
- age (peak 6 -15 months)
- Gender (m>f)
- race
- genetic background and socioeconomic status
- breast milk feeding
- degree of exposure to tobacco smoke
- degree of exposure to other children
- presence or absence of respiratory allergy
- season of the yr
- vaccination status
- Certain types of congenital craniofacial anomalies are particularly prone to OM

Etiologies

- Usually complicate URTI (Viral/bacterial)

Acute otitis media

- Streptococcus pneumoniae
- Nontypeable Haemophilus influenzae (conjunctivitis + OM)
- Moraxella catarrhalis

Otitis Media with Effusion

- Only 30 % culture positive
- bacterial DNA and viral RNA in much larger proportions

Clinical manifestation

- variable, especially in infants and young children.
 - Irritability or a change in sleeping or eating habits and occasionally, holding or tugging at the ear.
 - Vomiting
 - Fever
 - Rupture of the tympanic membrane with purulent otorrhea (uncommon)
 - Systemic symptoms and symptoms of URTI
 - Asymptomatic disease discovered at a routine health examination.
- OME often is not accompanied by overt complaints but can be accompanied by hearing loss.

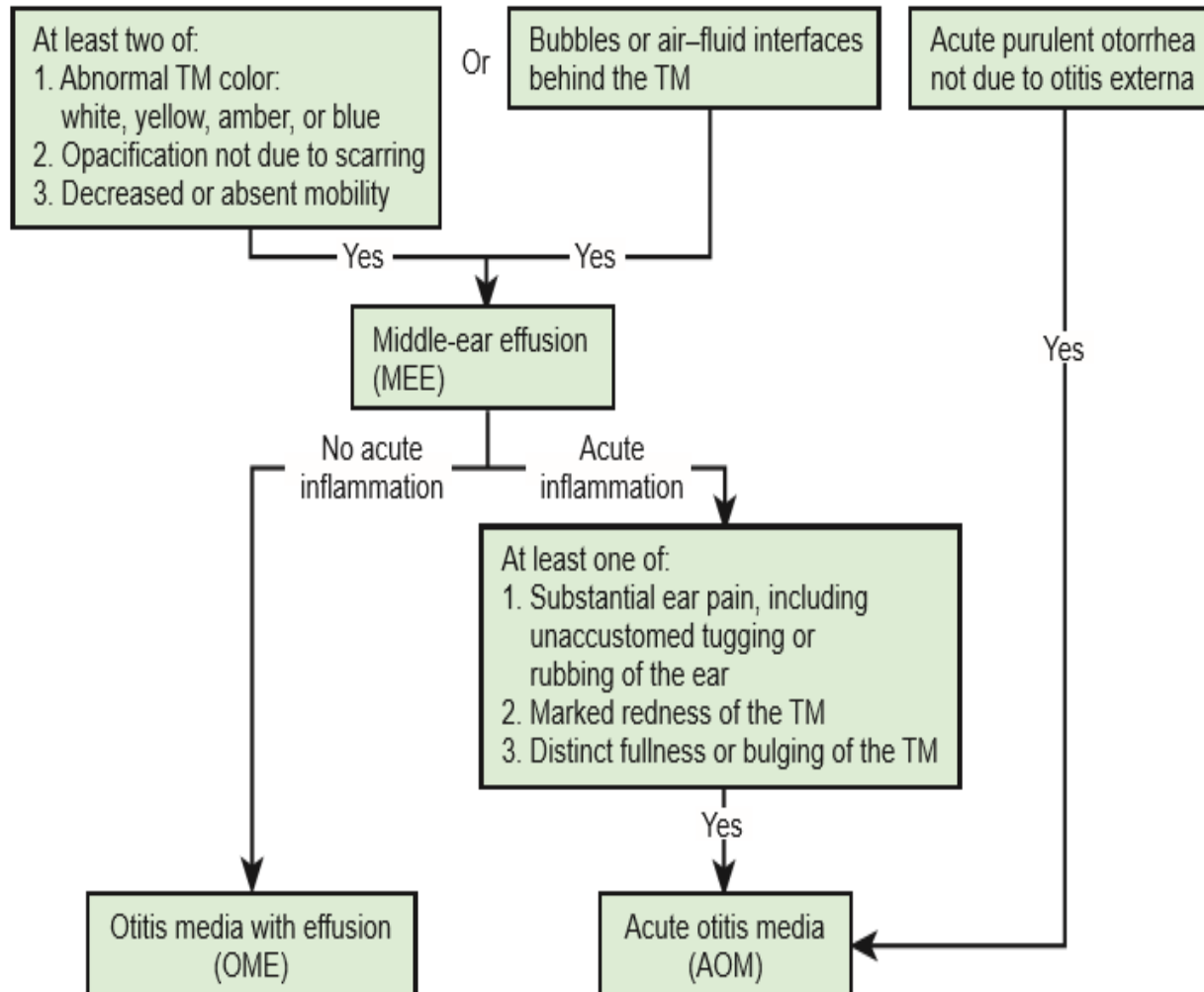
Dx

- Otoscopy
 - Moderate to severe bulging of the TM or new-onset otorrhea not caused by otitis externa
 - Mild bulging of the TM and recent (<48 hr)



n or intense TM
 f AOM should not
 : MEE
 ry : dx MEE

- Of all the visible characteristics of the TM, mobility is the most sensitive and specific in determining the presence or absence of MEE
- Bulging of the TM is the most specific finding of AOM (97%) but has lower specificity (51%)
- Otitis media with effusion (OME) is defined as the presence of middle-ear effusion (MEE) in the absence of acute signs of infection.
- In OME, bulging of the TM is absent or slight or the membrane may be retracted (Fig. 640-4); erythema also is absent or slight, but may increase with crying or with superficial trauma to the external auditory canal incurred in clearing the canal of cerumen.
- tympanogram provides information about TM compliance



Algorithm for distinguishing between acute otitis media and otitis media with effusion. TM, tympanic membrane

TREATMENT

- Supportive care: pain and fever management
- Antibiotics ; 10 days course
 - 1st line: Amoxicillin/ amox-clavulunate
 - 2nd line : Cephalosporines
- Myringotomy and Tympanocentesis

Complications

- **Intratemporal** : dermatitis, TM perforation, chronic suppurative OM, mastoiditis, hearing loss, facial nerve paralysis, cholesteatoma formation, and labyrinthitis
- **Intracranial** : Meningitis, epidural abscess, subdural abscess, focal encephalitis, brain abscess, sigmoid (lateral) sinus thrombosis, and otitic hydrocephalus

ACUTE OTITIS MEDIA (OM)

- Suppurative infection of the middle ear cavity
- The common bacterial pathogens are
 - S.pneumoniae
 - Nontypable H.influenzae
 - M.catarrhalis,
 - less frequently GAS
- Peak incidence of acute otitis media is in the second 6 months of life
- The major **risk factors** for acute OM are
 - Congenital Anomalies
 - Young age
 - No breast feeding
 - Parental history of ear infection, the presence of a sibling in the home (especially a sibling with a history of ear infection).
 - Passive exposure to tobacco smoke from parental smoking .

OTITIS MEDIA CONT...

CLINICAL MANIFESTATIONS

- In **infants** the frequent symptoms of acute OM are nonspecific & include fever, irritability, and poor feeding
- **Older children and adolescents** ,acute OM usually in association with fever & **otalgia** (acute ear pain)
- Acute OM also may present with **otorrhea**, or ear drainage after spontaneous rupture of tympanic membrane

Investigation

- Otoscopic examination
- Tympanometry

OTITIS MEDIA CONT...

TREATMENT

- Amoxicillin
80-90mg/kg po tid for
10 days
- Amoxicillin-
clavulanate 80-90mg/kg
if above treatment fails

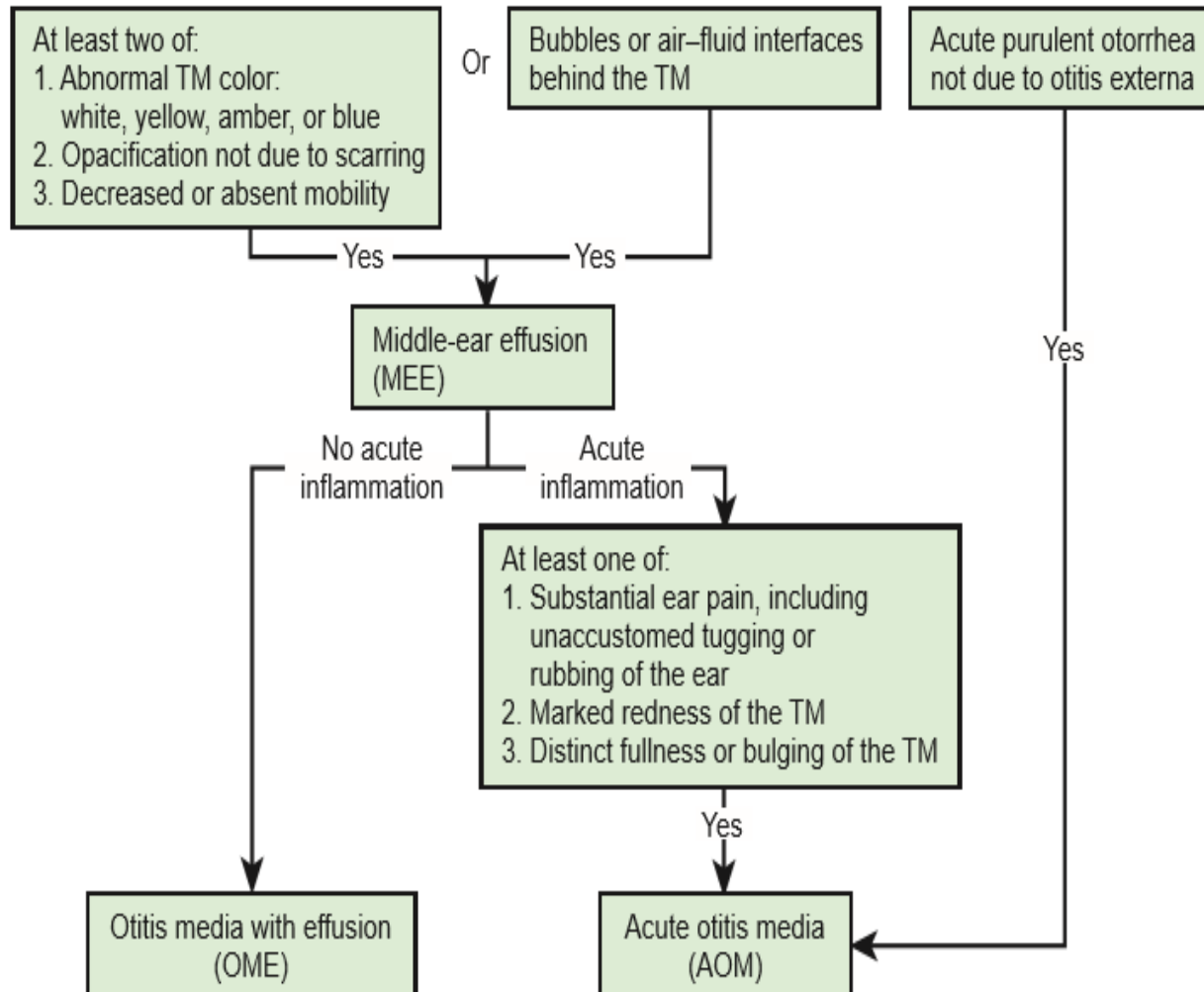
COMPLICATION OF AOM CONT...

INTRACRANIAL COMPLICATIONS

- Meningitis, epidural abscess, subdural abscess, focal encephalitis, brain abscess
- Sigmoid sinus thrombosis (also called *lateral sinus thrombosis*), and
- *Otitic hydrocephalus*

Intratemporal Complications

- Dermatitis
- TM perforation
- Chronic suppurative OM (CSOM)
- Acute mastoiditis
- Hearing loss
- Facial nerve paralysis
- Cholesteatoma formation- is an abnormal accumulation of squamous epithelium within the middle ear and mastoid.
- Labyrinthitis



Algorithm for distinguishing between acute otitis media and otitis media with effusion. TM, tympanic membrane

TREATMENT

- Supportive care: pain and fever management
- Antibiotics ; 10 days course
 - 1st line: Amoxicillin/ amox-clavulunate
 - 2nd line : Cephalosporines
- Myringotomy and Tympanocentesis